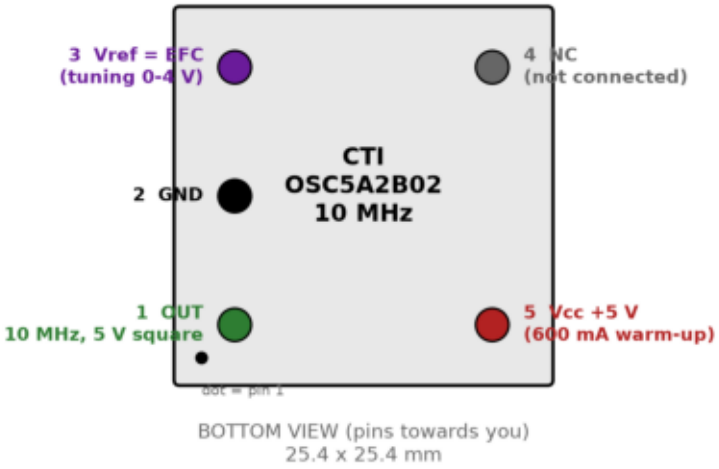
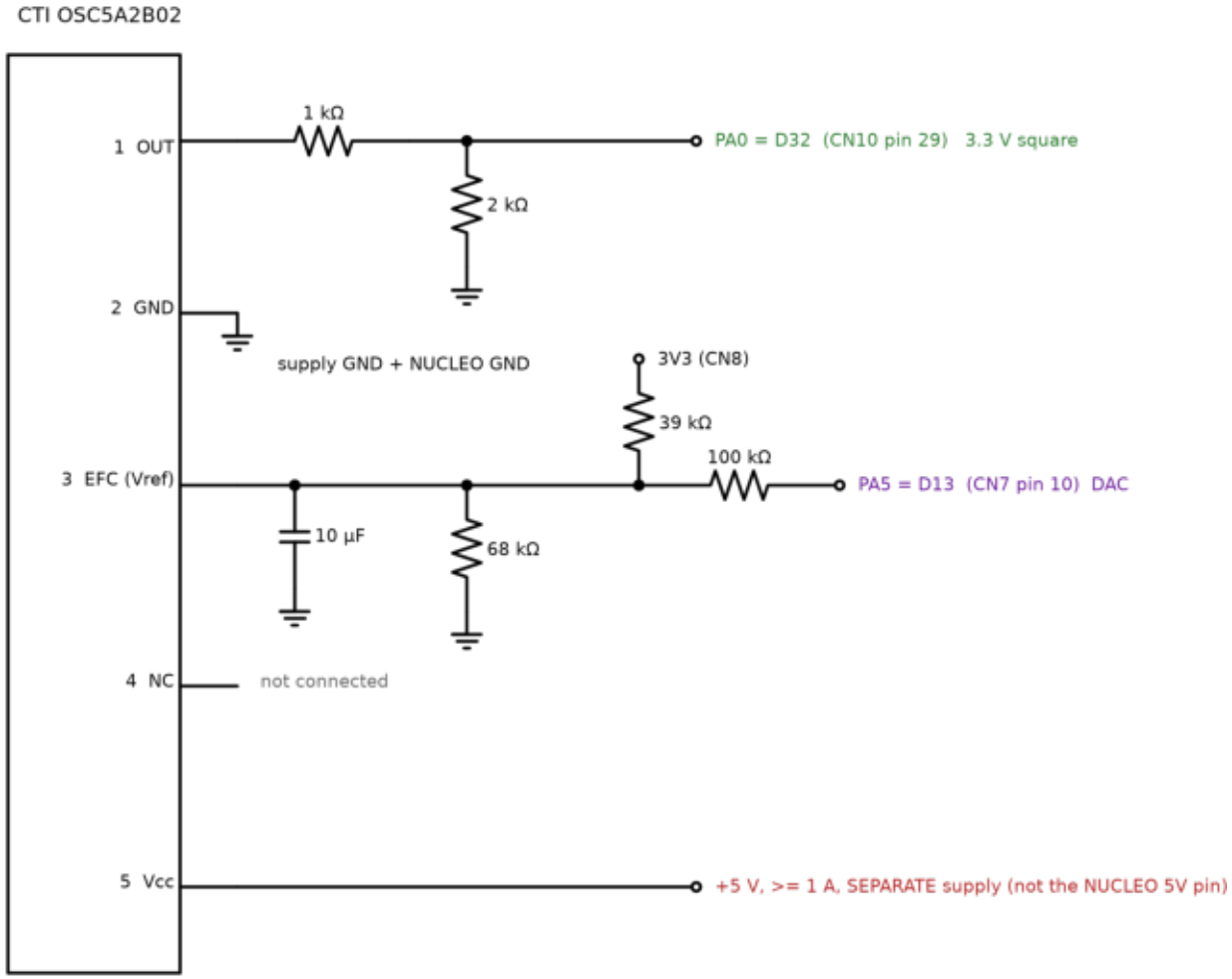
OCXO test on the NUCLEO-H723ZG: CTI OSC5A2B02 wiring

Firmware build -DOCXO_REF (commit 9eab67b): TIM2 counts the OCXO on PA0, TIM3 captures every 2500 counts; DAC on PA5 tunes it.





1. Power: pin 5 to a separate 5 V supply (>= 1 A: 600 mA during the first minutes, 250 mA warm). Pin 2 to that supply's GND AND to NUCLEO GND.
2. Clock: pin 1 (5 V square) -> 1 kΩ -> PA0 / D32, and 2 kΩ from PA0 to GND (5 V -> 3.3 V). Keep it short (< 10 cm), twist it with a ground wire.
3. Tuning: pin 3 ('Vref' on the datasheet = the frequency control input). Network: 39 kΩ from 3V3, 68 kΩ to GND, 100 kΩ from PA5 / D13, 10 µF to GND.
   PA5 open or DAC mid-scale: about 2.0 V = the OCXO's nominal centre. DAC 0 ... 3.3 V -> about 1.67 ... 2.33 V (about +-0.25 ppm).
4. Pin 4: not connected. Pin 1 is marked by the dot (bottom view: pins towards you, pin 1 bottom left, pin 5 bottom right).
5. Warm up >= 10 min on its own supply, then the OCXO build is flashed. 'OCXO reference OK' in the log = captures at 4 kHz found.